Assignment 1 (Due: Nov. 8, 2026)

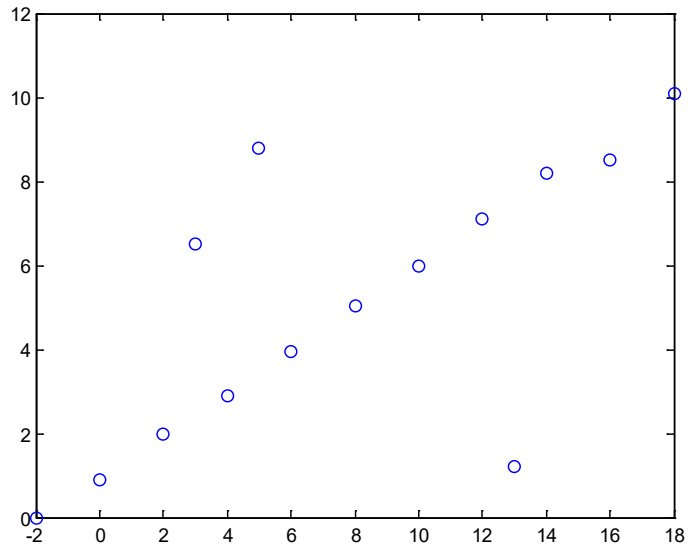
1. (**Math**) In our lectures, we mentioned that matrices that can represent Euclidean transformations can form a group. Specifically, in 3D space, the set comprising matrices $\{\boldsymbol{M}_i\}$ is actually a group, where
   $\boldsymbol{M}_i = \begin{bmatrix} \boldsymbol{R}_i & \boldsymbol{t}_i \\ \boldsymbol{0}^T & 1 \end{bmatrix} \in \mathbb{R}^{4\times4}$, $R_i \in \mathbb{R}^{3\times3}$ is an orthonormal matrix, det($\boldsymbol{R}_i$)=1, and $\boldsymbol{t}_i \in \mathbb{R}^{3\times1}$ is a vector.
   Please prove that the set $\{\boldsymbol{M}_i\}$ forms a group.
   Hint: You need to prove that $\{\boldsymbol{M}_i\}$ satisfies the four properties of a group, i.e., the closure, the associativity, the existence of an identity element, and the existence of an inverse element for each group element.

2. (**Math**) When deriving the Harris corner detector, we get the following matrix $\boldsymbol{M}$ composed of first-order partial derivatives in a local image patch w,
   $$\boldsymbol{M} = \begin{bmatrix} \sum_{(x_i,y_i)\in w} (I_x)^2 & \sum_{(x_i,y_i)\in w} (I_x I_y) \\ \sum_{(x_i,y_i)\in w} (I_x I_y) & \sum_{(x_i,y_i)\in w} (I_y)^2 \end{bmatrix}$$
   a) Please prove that $\boldsymbol{M}$ is positive semi-definite.
   b) In practice, $\boldsymbol{M}$ is usually positive definite. If $\boldsymbol{M}$ is positive definite, prove that in the Cartesian coordinate system, $[x, y]\boldsymbol{M}\begin{bmatrix} x \\ y \end{bmatrix} = 1$ represents an ellipse.
   c) Suppose that $\boldsymbol{M}$ is positive definite and its two eigen-values are λ_1 and λ_2 and $\lambda_1 > \lambda_2 > 0$. For the ellipse defined by $[x, y]\boldsymbol{M}\begin{bmatrix} x \\ y \end{bmatrix} = 1$, prove that the length of its semi-major axis is $\frac{1}{\sqrt{\lambda_2}}$ while the length of its semi-minor axis is $\frac{1}{\sqrt{\lambda_1}}$.

3. (**Math**) In the lecture, we talked about the least square method to solve an over-determined linear system $\boldsymbol{Ax} = \boldsymbol{b}, \boldsymbol{A} \in \mathbb{R}^{m\times n}, \boldsymbol{x} \in \mathbb{R}^{n\times1}, m > n, rank(\boldsymbol{A}) = n$. The closed form solution is $\boldsymbol{x} = \left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T\boldsymbol{b}$. Try to prove that $\boldsymbol{A}^T\boldsymbol{A}$ is non-singular (or in other words, it is invertible).

4. (**Programming**) RANSAC is widely used in fitting models from sample points with outliers. Please implement a program to fit a straight 2D line using RANSAC from the following sample points:

(-2, 0), (0, 0.9), (2, 2.0), (3, 6.5), (4, 2.9), (5, 8.8), (6, 3.95), (8, 5.03), (10, 5.97), (12, 7.1), (13, 1.2), (14, 8.2), (16, 8.5) (18, 10.1). Please show your result graphically.



5. (**Programming**) Get two images I_1 and I_2 of our campus and make sure that the major parts of I_1 and I_2 are from the same physical plane. Stitch I_1 and I_2 together to get a panorama view. There are additional requirements:
   1) The program should be implemented with C++ and OpenCV;
   2) For key point detection and matching, ORB features should be used;
   3) For solving linear equation systems, Moore-Penrose inverse should be explicitly used.